Ps 02

Erik An

<2026-01-26 Mon>

Contents

1 Problem 1 **DONE** **2**

1.1 Task 2

1.2 Solution 2

2 Problem 2 **DONE** **2**

2.1 Task 2

2.2 Solution 2

3 Problem 3 **DONE** **3**

3.1 Task 3

3.2 Solution 3

4 Problem 4 **4**

4.1 Task 4

4.2 Notes 4

4.3 Solution 4

5 Problem 5 **DONE** **4**

5.1 Task 4

5.2 Solution 4

1 Problem 1 DONE

1.1 Task

Show that the function $f : \mathbb{Q} \to \mathbb{R}$ given by $f(x) = \begin{cases} 0, & x^2 < 2, \\ 1, & x^2 > 2, \end{cases}$ is continuous.

1.2 Solution

Let $c \in \mathbb{Q}$. We know that $c^2 \neq 2$ as $\sqrt{2} \notin \mathbb{Q}$. Hence, we have 2 cases:

- Case 1: $c^2 < 2$
  Choose $\delta < \sqrt{2} - |c|$.
  Hence, $\forall x \in (c - \delta, c + \delta), |x| < \sqrt{2}$.
  Thus, $\lim_{x \to c} f(x) = 0 = f(c)$.
  Hence, f is continuous $\forall x : x^2 < 2$.
- Case 2: $c^2 > 2$
  Choose $\delta < |c| - \sqrt{2}$.
  Hence, $\forall x \in (c - \delta, c + \delta), |x| > \sqrt{2}$.
  Thus, $\lim_{x \to c} f(x) = 1 = f(c)$.
  Hence, f is continuous $\forall x : x^2 > 2$.

Thus, $f : \mathbb{Q} \to \mathbb{R}$ is continious.

2 Problem 2 DONE

2.1 Task

Prove that the following functions are continuous: (a) $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \frac{1}{x^2+1}$ (b) $g : \mathbb{R} \to \mathbb{R}$, $g(x) = \exp(|x|)$.

2.2 Solution

- Part (a)
  A function $p(x) = x$ is known to be continuous on $x \in \mathbb{R}$.
  - x^2 is a product of two continuous functions p.

- 1 is a continuous function.
- $x^2 + 1$ is a sum of two continuous functions.
- Since $x^2 + 1 \geq 1 \, \forall x \in \mathbb{R}$, $\frac{1}{x^2+1}$ is well defined $\forall x \in \mathbb{R}$.

Thus, a function $f(x) = \frac{1}{x^2+1}$ is continuous on $\mathbb{R}$.

- Part (b)
  - The absolute value function $h(x) = |x|$ is known to be continuous on $x \in \mathbb{R}$.
  - The exponential function $r(x) = e^x$ is also known to be continuous on $x \in \mathbb{R}$.

  The function $g(x) = e^{|x|}$ is a composite function $g = r \circ h$, and hence continuous in $\mathbb{R}$.

3 Problem 3 DONE

3.1 Task

Prove or disprove that the function $f : \mathbb{R} \to \mathbb{R}$ is continuous, where f is given by $f(x) = \begin{cases} \exp(-\frac{1}{x}), & x \in \mathbb{R} \backslash \{0\}, \\ 0, & x = 0. \end{cases}$

3.2 Solution

To determine continuity at $x = 0$, I examine the left and right-hand limits of $f(x)$:

- Left-Hand Limit ($x \to 0^-$) Let $t = -\frac{1}{x}$. As $x \to 0^-$, I know that $\frac{1}{x} \to -\infty$, hence, $t \to \infty$.

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} e^t = \infty$$

- Right-Hand Limit ($x \to 0^+$) Let $t = -\frac{1}{x}$. As $x \to 0^+$, I know that $\frac{1}{x} \to \infty$, hence, $t \to -\infty$.

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} e^t = 0$$

Conclusion: Since $\lim_{x \to 0^-} e^t = \infty$ the limit $\lim_{x \to 0} f(x)$ does not exist. Therefore, $f(x)$ is discontinuous at $x = 0$. Because limit is infinite, this is classified as infinite discontinuity.

4 Problem 4

4.1 Task

Write a Julia function interpol(X, Y) with inputs $X = (x_1, ..., x_n)^T, Y = (y_1, ..., y_n)^\top \in \mathbb{R}^n$ and the output is a function f such that

(a) $f(x_i) = y_i$, for $i = 1, ..., n$ and f is continuous (find a simple function that satisfies those criteria). (b) $f(x_i) = y_i$, for $i = 1, ..., n$ and f is a step-function.

4.2 Notes

4.3 Solution

```
nil
```

5 Problem 5 DONE

5.1 Task

Scientists study how global mean sea-level rise $S(t)$ (in centimeters) depends on the global temperature anomaly t, which measures how many degrees Celsius the current global average temperature differs from a chosen historical baseline. A simplified physical model predicts

$$S(t) = \frac{3}{2} + \frac{1}{2}t + 2(\exp(\frac{2}{5}t) - 1)$$

where the constant $\frac{3}{2}$ cm accounts for committed sea-level rise from past emissions, the linear term $\frac{1}{2}t$ represents thermal expansion of seawater, and the exponential term $2(\exp(\frac{2}{5}t) - 1)$ models nonlinear ice-sheet melt feedbacks.

The Paris International Climate Treaty defines a critical threshold of 8 cm sea-level rise that triggers coastal emergency measures.

(a) Implement a bisection method in Julia or familiarize yourself with the code provided. (b) Use the implementation to determine the critical global temperature anomaly, that is, determine t such that $S(t) = 8$.

5.2 Solution

- Part (a)

  Note: While not explicitly required, I will first verify that the method is applicable.

To apply the *Bisection Method* for finding a root, I must first prove that the function is continuous $\forall t \in \mathbb{R}$. If the function were discontinuous, the Bisection Method (and the Intermediate Value Theorem it relies on) might fail.

Let's look at the given function:

$$S(t) = \frac{3}{2} + \frac{1}{2}t + 2(e^{\frac{2}{5}t} - 1)$$

Expanding the terms:

$$\begin{aligned} S(t) &= \frac{3}{2} + \frac{1}{2}t + 2e^{\frac{2}{5}t} - 2 \\ &= 2e^{\frac{2}{5}t} + \frac{1}{2}t - \frac{1}{2} \end{aligned}$$

We can view $S(t)$ as the sum of two elementary functions, $r(t)$ and $h(t)$:

$$\begin{aligned} r(t) &= 2e^{\frac{2}{5}t} \quad \text{(Exponential)} \\ h(t) &= \frac{1}{2}t - \frac{1}{2} \quad \text{(Linear Polynomial)} \end{aligned}$$

Both functions and polynomials are known to be continuous for all real numbers. Since the sum of continuous functions is also continuous, $S(t)$ is continuous $\forall t \in \mathbb{R}$.

Therefore, the Intermediate Value Theorem (IVT) applies:

If f is continuous on $[a, b]$ and $f(a) < 0$ and $f(b) > 0$, then $\exists \xi \in (a, b) : f(\xi) = 0$.

Since continuity is established, the Bisection Method is a valid choice.

```
function bisection_method(f, a, b, tol=1e-6)
    m = 0.0
    while (abs(a-b) > tol)
        m = (a+b)/2
        (f(a) * f(m) < 0) ? b = m : a = m
    end
    return m
end

S(t) = 3/2 + t/2 + 2*(exp((2t)/(5))-1)

bisection_method(S, -10, 10)
```

```
-1.3401728868484497
```

- Part (b)

```
function bisection_method(f, a, b, tol=1e-6)
    m = 0.0
    while (abs(a-b) > tol)
        m = (a+b)/2
        (f(a) * f(m) < 0) ? b = m : a = m
    end
    return m
end

S(t) = 3/2 + t/2 + 2*(exp((2t)/(5))-1)

root = bisection_method(t -> S(t) - 8, -10, 10)

println("Critical Temperature Anomaly: $root")
```

```
bisection_method (generic function with 2 methods)
S (generic function with 1 method)
3.111853003501892
Critical Temperature Anomaly: 3.111853003501892
```